BrickPlan: Diagnosing Future Action Cost and Error Recovery in Visual Assembly Planning

Benchmark research release 2.2
Population model and human studies pending

Abstract

Visual assembly planning requires predicting how an action changes later access, workspace occupancy, and unfinished work. Final-shape accuracy alone does not isolate these abilities. We introduce BrickPlan, an executable benchmark with exact future-cost and recovery labels under a finite assembly abstraction. Original LDraw geometry determines qualified motions and connectivity; a limited rack and persistent off-assembly inspection ledger couple decisions over time. The release contains 377 interactive episodes and 1,952 diagnostic questions from 16 source groups. Matched sequences test future legality, fixed-context contrasts test geometric dependence, and delayed-repair questions exclude the goal from recovery credit. Static diagnostics supply four fixed views; active diagnostics charge additional acquisitions. Calibration questions and repeated variants receive explicit scoring treatment. Structural controls reveal both scope and limits: closure greedy completes 1,061 of 1,131 task-runs but only 873 optimally; an inverse-last-action control solves 58.0% of recovery questions. We release native-resolution observations, exact evaluators, a question workbench, and a preserved historical model trace. These measurements characterize the instrument; population visual-model and human performance remain unmeasured.

1. Introduction

An assembly action can be legal now and expensive later. Removing a piece may consume the last available holder before a deeper part becomes accessible; replacing another may obstruct unfinished inspection work. If an agent detects an error after several further actions, reversing its latest action need not restore an efficient completion. Measuring these effects requires more than recognizing brick types or comparing a final image.

BrickPlan asks whether an agent can predict the future action cost of a visible assembly decision and recover after a costly choice has propagated. We distinguish immediate rejection, a legal action that increases exact remaining cost, and failure to return to an optimal completion corridor. Restricting placements to registered original sites makes these distinctions exhaustively labelable. This deliberate restriction trades manipulation breadth for exact diagnostic targets.

A preceding assembly-only bank illustrates the need for explicit controls. An occupied-blocker closure formula reproduced all 1,802,728 stored state–goal costs. Long solutions alone did not establish deep planning. BrickPlan adds finite external storage and inspection work whose completion persists after reattachment. Static relations remain useful, but occupancy alone no longer specifies the mission state or exact cost.

Our contributions are threefold. First, we define an executable visual assembly task with resource and work obligations, interactive episodes, and explicit checkpoint diagnostics. Second, we construct sequence, geometry, appearance, and recovery controls, with separate static and active observations and auditable scoring units. Third, we release the construction, execution, evaluation, and publication pipeline, including strong structural controls. We do not infer a human–model gap, required internal reasoning depth, or broad physical competence from software verification.

2. Evaluation design

Table 1 fixes the reporting framework before empirical collection. Primary episode outcomes are completion and optimal completion over all assignments, plus excess operations among successes. Diagnostic accuracy is reported by family, scoring role, canonical problem, and source. Joint pair scores require both members correct. Invalid outputs and missing assignments remain in their denominators. Privileged structural policies are separate from image-input results.

The study compares matched assignments and identity seeds with declared action and view budgets. For each VLM, static image and structured conditions share the same initial views, executor, and answer syn-

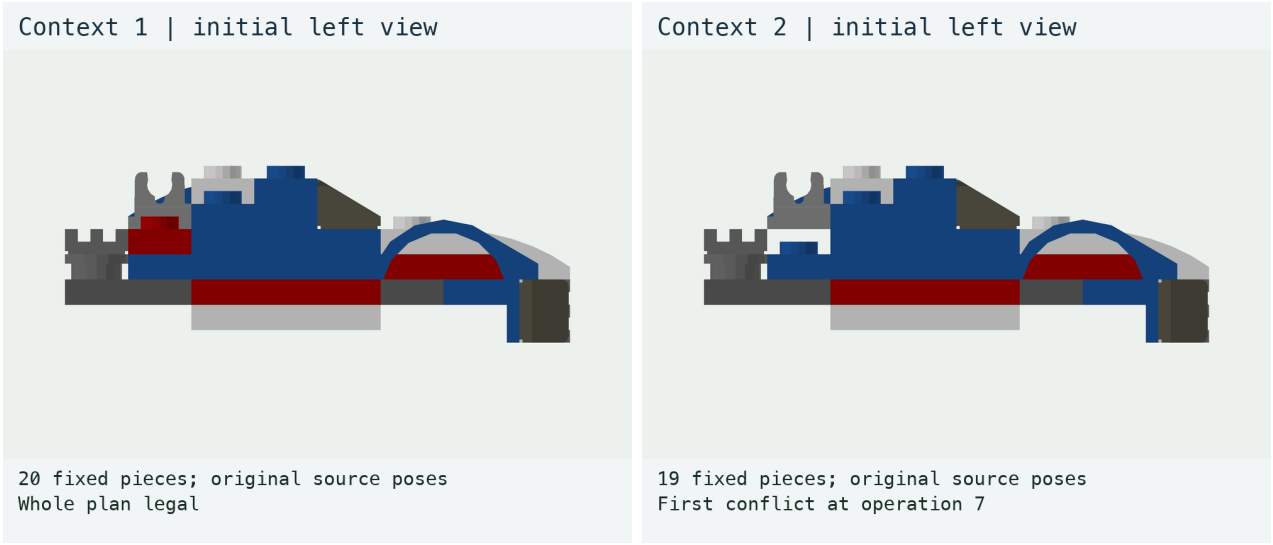


Figure 1. An answer-changing geometry pair shown from the initial left view, supplied before answering in the static condition. Registered pieces, original poses, occupancy, target, rack, work orders, and proposed plan are matched. The fixed context differs by one original source piece. Context 1 permits the whole plan; context 2 rejects operation 7 after six shared legal steps. Their initial isometric views are identical; the side view reveals the difference. Pixel distinction does not by itself certify human or model answerability. The unequal fixed-part counts remain a confound.

Table 1. Experiment framework. Population results are unmeasured; model snapshots and provider settings must be fixed before collection.

Condition	Evidence status
Structural policies and exact search	Measured
SmolVLM-256M, image input	Historical failed case
Open / proprietary VLMs	Pending
Humans, images + registered IDs	Pending
Static / active / structured input	Matched study planned
Stepwise / explicit-plan prompting	Matched study planned

tax; active acquisition is reported separately. A format/cost pilot precedes the full study. Report snapshot, provider, prompt, image preprocessing, context policy, decoding, calls, measured token use, and acquisition count. Human collection requires documented instructions, training, participant counts, and consent before recruitment.

Identity seeds, paired questions, and variants reuse source geometry. Per-source results and coverage precede pooled claims. Only eight source groups are held out from development, limiting transfer uncertainty. More diagnostic rows do not create more independent assemblies. Errors are stratified into parsing, immediate legality, future cost, diagnosis, and repair.

3. Related work

LTRON's Break and Make evaluates interactive structural understanding through disassembly and reconstruction [5]. InstructioNet studies construction supported by an agent's own explicit visual instructions [6]. These establish interactive brick construction and long sequences; BrickPlan does not claim either as new. Its registered-site action space permits exact counterfactual completion costs and event-level recovery labels.

LEGO-Puzzles includes both elementary visual questions and a Planning set [4]; it should not be reduced to recognition. PhyBlock combines hierarchical assembly and physical-understanding questions [3]. Brick-Composer introduces BC-Bench for diverse brick selection and pose estimation [2]. BrickPlan fixes placement sites and instead measures resource coupling, future legality, and delayed corridor return. It complements broader spatial tasks.

Shortest-path search and admissible heuristics are established tools [1]. Failure of a weak policy does not prove hardness. We therefore include a strong static relaxation, capacity-aware dispatch, bounded and full A*, and shallow-choice bounds. The contribution is an auditable visual diagnostic instrument, not a new planning algorithm.

4. Task and observation contract

4.1. State, geometry, and actions

Let $A \subseteq V$ be occupied registered sites, $G \subseteq V$ the target, $J \subseteq V$ required inspections, and $D \subseteq J$ completed inspections. The mission state is $x = (A, D)$. Each piece P_i belongs to one original site S_i. Detached pieces occupy universal oversized rack holders, subject to

$$r(x) = |V \setminus A| \leq K. \tag{1}$$

An audit completes a pending job only while its piece is detached; completion persists after replacement. The terminal condition is $A = G,\ D = J$, followed by an explicit finish.

An attach or detach must satisfy qualified straight translation, collision, stud-connectivity, registration, and capacity rules. Fixed pieces form a connected clamped assembly. Executable positions preserve original LDraw transforms. Guarded inset meshes and prescribed paths define qualification; this is not force, gravity, clutch, hand-access, or arbitrary rigid-body simulation. Rack transfer is a resource abstraction. Exploded views are reference diagrams, never executable states.

Each physical operation or audit costs one operation; so does a rejected operation attempt. Camera acquisition and diagnosis have separate budgets. Episodes allow 12 additional views and 12 diagnoses. Public operation budgets depend on visible problem size and work, not private optimum cost.

4.2. Matched observations

Five raster panels show the current assembly, target, all-site reference, registered-piece catalog, and rack. Main views are supersampled at 3840×2880 and reduced to native 1920×1440; catalog tiles are 480×360. Text gives P/S registration, rack inventory, and the inspection ledger. Image input omits pose matrices, installed-site flags, legal-action lists, relation graphs, and exact costs. Explicit registration simplifies correspondence without supplying geometric rules.

Static checkpoint diagnostics provide isometric, left, rear, and right views at revision zero, with later acquisition disabled. Active image diagnostics start with isometric only and charge one acquisition per inspect. Structured diagnostics use the same four initial views and additionally provide original matrices, colors, part types, path directions and lengths, and occupancy. They omit the private blocker/support graph; structural software controls receive that graph. Interactive episodes retain their common active-view protocol.

Acquired frames persist with their revision. Free 3D orbit, explosion, and answer review are available in unscored practice only. Human and model policies share the controlled observation and submission contract.

4.3. Diagnostic formats

Table 2 separates interactive execution from checkpoint questions. There are 653 next-action questions, 288 ordinary sequence questions, 20 geometry questions, 48 appearance questions, and 943 delayed-repair questions. Sequence questions expose the complete proposed plan and ask for its first illegal operation, or null. Recovery questions expose the pre-error checkpoint, executed history, current observation, and legal candidates. They ask for a shortest repair step, not hidden model reasoning.

5. Construction and validity controls

5.1. Source groups and sampling

The 99 qualified geometry profiles include 17 base assemblies and 82 context profiles, including zero-yield candidates. They belong to 16 source kit groups. Alternate builds and context variants remain within one group: roadster and kart both belong to 31027. There are 109 development episodes from eight groups and 268 held-out-source episodes from eight other groups. Of the latter, 256 follow seeded source sampling and 12 are mechanism-conditioned geometry episodes.

Version 2.2 preserves all 145 earlier episodes and 739 diagnostic records, adding 232 episodes and 1,213 questions. Of 11 additional frozen candidates, five pass admission: 10036, 10213, 10242, 10265, and 10269. Six rejected candidates remain documented. Some admitted subassemblies have only two or four registered sites, so source growth is not equated with difficult geometry.

Legacy tasks used mechanism and solver evidence during selection. Additional candidates and seeded sampling were recorded before their solver evaluation. Initial collision failures motivated a disclosed geometry-only admission refinement: remove conflicting instances in deterministic source order and retain a stud-connected component. It moves no parts. After qualification, admission uses solvability and seeded sampling, not observed policy difficulty. Initial failures, retained indices, and qualification results are archived. "New source" refers to benchmark construction, not absence from pretraining.

5.2. Exact costs and difficulty descriptors

For each mission, we enumerate the capacity-constrained physical component and inspection-bit product. Reverse shortest paths give $C^*(x)$, the minimum remaining operations. The 130 tables contain 6,836,215 state entries and 72,954,915 directed-

Table 2. Seven public question families. Episode families are mutually exclusive presentation groups. Every entry has an ID, prompt, constraints, answer format, and scoring rule.

Family	Question	Response and measurement
Assembly / reconfiguration	Transform current assembly into target.	Actions; completion and excess cost
Finite-resource scheduling	Free the holders required for later work.	Actions under a capacity bound
Inspection / persistent state	Complete off-assembly work and restore target.	Actions and audits; physical and ledger completion
Next-action judgment	Which legal choice minimizes completion cost?	Index; any optimum accepted
Future-sequence checking	Where is the first illegal operation?	One-based index or null
Geometry / appearance	Does context change, or recoloring preserve, legality?	Conflict answer and joint pair score
Delayed error recovery	Which choice returns to the earlier optimal corridor?	Index on a shortest nonterminal repair

Table 3. Fixed operation-length bins: D1 (1–8), D2 (9–16), D3 (17–24), D4 (25+). They are not cognitive-depth levels.

Tier	Episodes	C^* range	Gap range	A* max
D1	97	1–8	0–0	8
D2	152	9–16	0–4	311
D3	83	17–24	0–10	4372
D4	45	25–42	0–18	12578

transition entries, with physical states repeated under different goals and jobs. Checks cover runtime transitions, physical inverses, lower bounds, and Bellman equalities.

Let R be occupied blocker closure needed to remove unwanted pieces, insert target pieces, and detach unfinished inspection pieces. A static relaxation is

$$h(x) = |A \triangle G| + 2|R \cap A \cap G| + |J \setminus D|. \qquad (2)$$

Each mismatch needs an operation, each occupied target piece in the required closure needs removal and replacement, and each pending job needs an audit. Capacity and connectivity can add operations. The maximum retained gap $C^* - h$ is 18. Registered sites, rack size, occupancy, jobs, gap, and search effort are separate descriptors; length bins alone do not imply planning difficulty.

5.3. Sequence and shallow-choice controls

The 144 legal/corrupted sequence pairs match first and last actions, action multisets, and prefixes before the first conflict. Legal witnesses contain no repeated-state padding. Each conflict bin (2–5, 6–10, 11+) has 48 pairs. A prefix-only control executes k operations and otherwise answers "all legal." It scores 158/288 at $k = 2$ and 192/288 at $k = 5$; zero-step accuracy is 50%. This reduces a measured predecessor shortcut without establishing immunity to other ones.

All 653 next-action checkpoints include an optimal choice in the depth-one closure minimizer set. Uniform tie selection yields 94.0% expected accuracy; uniform selection among all candidates yields 36.9%. This strong result uses privileged structural relations. We disclose tie sizes, optimistic and pessimistic accuracy, and the matched image/structured comparison rather than infer required reasoning depth.

5.4. Geometry and appearance

Ten answer-changing pairs cover four interventions and three groups (31027, 10014, 10036). They match registered pieces and poses, start, target, rack, work, and plan; induced fixed contexts are independently qualified. Of 41 qualified context pairs, 37 produce no retained future contrast under the declared search. Every retained pair has unequal fixed-part counts. They are not count-matched counterfactuals.

Native certificates compare all five panels in four initial views. Every pair differs in a supplied static view; three pairs have identical isometric inputs alone (Figure 1). This establishes pixel distinction, not perceptual sufficiency or successful acquisition by an active agent. Twenty-four original/grayscale pairs preserve geometry, state, plan, and answer. These measure appearance invariance, not additional independent structures.

5.5. Delayed repair and calibration

For a pre-error checkpoint b, let $\mathcal{C}_b^-$ contain all nonterminal states on any optimal completion from b. After a legal positive-excess error, a distinct-state legal his-

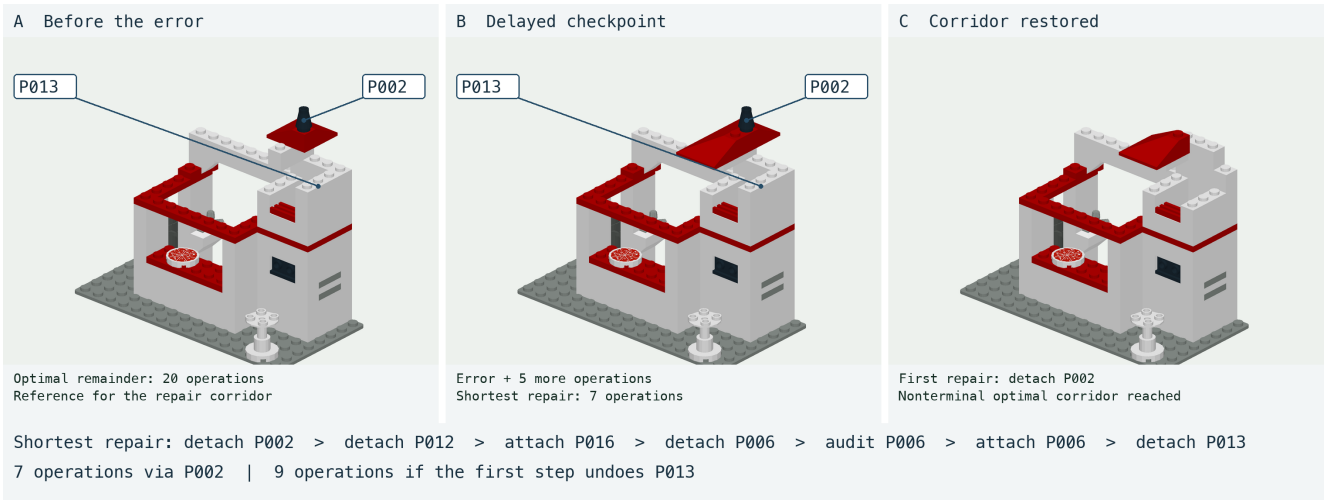


Figure 2. A certified lag-five case (source-10036-35/repair-lag-5). A: the pre-error checkpoint defines an optimal completion corridor. B: an error and five further operations leave it. The last history action attached P013. Immediately undoing it requires nine operations to repair; the optimum takes seven, starting with detaching P002. C: that shortest repair reaches a nonterminal corridor state. These are oracle replay frames, not model reasoning or a model success. Exact states and actions are archived with figure provenance.

tory of lag 1, 3, or 5 remains outside this corridor. At checkpoint z, the repair target is

$$r_b(z) = \min_{y \in \mathcal{C}_b^-} \operatorname{dist}(z, y). \tag{3}$$

The 943 questions have repair costs 1–7, independently verified by forward BFS. Any first action on a shortest repair is accepted. Finishing after a costly detour does not itself earn repair credit.

Lag alone does not establish nonlocal repair. Reversing the last public action solves 547/943 questions (58.0%); the inverse is suboptimal on 368 and unavailable on 28. These cohorts are reported separately. All 26 questions with every candidate optimal remain calibration checks and receive no weight in discriminative accuracy.

6. Metrics and scoring units

For a legal unit-cost action with successor x', conditional excess is

$$\delta(x, a) = 1 + C^*(x') - C^*(x) \geq 0. \tag{4}$$

A rejected attempt leaves state unchanged and contributes one excess operation. Thus legal but costly actions remain distinct from immediate invalidity even when rework was already mandatory. Successful regret is $T - C^*(x_0)$ for T charged attempts. Completion and optimal completion use all assignments; mean successful regret accompanies its coverage.

Every legal positive-excess action opens a recovery event. Later errors do not overwrite earlier events. Restoration, immediate and lag-specific cohorts, and right-censored unfinished trajectories are recorded separately. Diagnosis concerns avoidable errors since the last diagnosis; mandatory rework is not an error. Suffix effort is compared with exact remaining cost at diagnosis. Survival curves preserve the first cost-loss event after later recovery, with assigned and unknown counts at each step. The reversible physical components and persistent audits contain no genuine irreversible dead ends; their eligible-choice rate is undefined.

The 1,952 rows represent 1,819 canonical diagnostic problems; repeated statements and appearance variants account for 133 excess rows. Canonical identity includes source geometry, state, target, work, capacity, and the exact question or history. There are 1,926 discriminative assignments. Primary accuracy averages assigned variants within a canonical problem, then problems within a source, then sources equally. Missing variants receive zero before averaging. Row, problem-macro, source-macro, calibration, and recovery-cohort scores remain available. Joint pair correctness requires both assigned members; a missing member earns zero. The supplement details metric denominators and regret conservation.

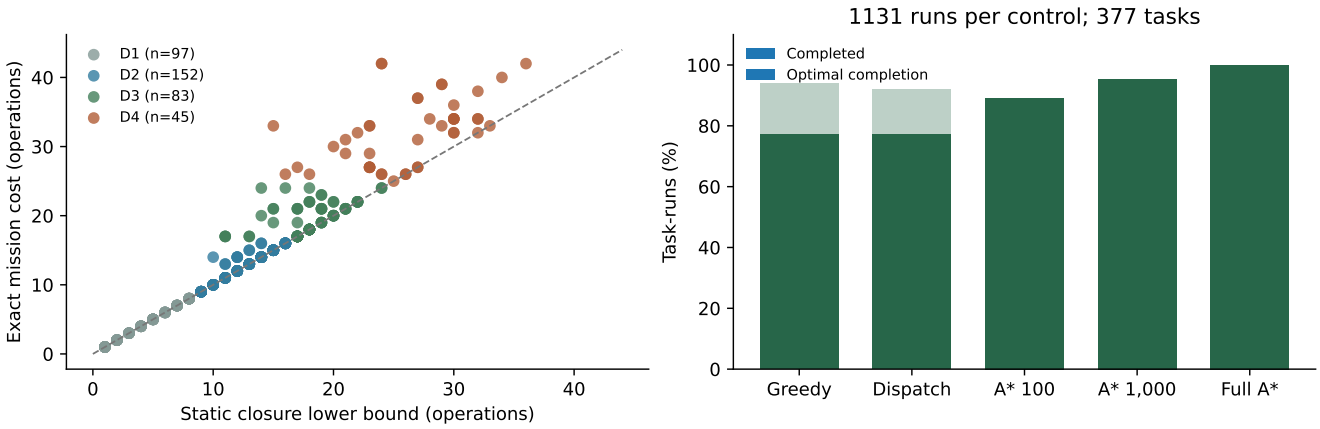


Figure 3. Measured software evidence. Left: exact cost and static lower bound, with coincident points overlapping. Right: completed and optimal task-runs differ for greedy controls. Length bins do not imply monotonic search difficulty. These summaries combine legacy development and seeded source samples; partition results are released separately.

Table 4. Structural controls on 377 episodes. The oracle uses one seed; other policies use three tie-order seeds (1,131 runs). Regret is mean excess among successes. Exact relations are privileged inputs, so these are not image-input VLM results.

Control	Runs	Success	Optimal	Regret	Max exp.
Exact oracle	377	377	377	0.00	—
Closure greedy	1131	1061	873	3.45	—
Capacity dispatch	1131	1039	873	3.45	—
Closure A* (100)	1131	1008	1008	0.00	100
Closure A* (1,000)	1131	1079	1079	0.00	1,000
Closure A* (full)	1131	1131	1131	0.00	12,578
Hamming A* (full)	1131	1131	1131	0.00	75,454
Legal random	1131	192	56	9.28	—
Drop inspections	1131	408	408	0.00	—
Ignore rack limit	1131	842	842	0.00	—

7. Measured evidence

7.1. Structural policies

Closure greedy minimizes the static successor bound with visited-state and randomized site tie-breaking. Capacity dispatch prioritizes audits, mandatory removals, safe placements, and temporary staging, with revisited-state escape. Greedy completes 1,061/1,131 runs and dispatch 1,039/1,131; both achieve 873 optimal completions. Completion alone therefore conceals avoidable effort (Table 4).

A* uses graph reopening, $(g + h, h)$ priority, and declared limits without oracle access. Limits of 100 and 1,000 expansions complete 1,008 and 1,079 runs. Full closure A* completes all 1,131 optimally, with median 14 and maximum 12,578 expansions. The weaker occupancy-plus-pending-audit heuristic is also reported. Strong perfect-information planning remains computationally modest; weak controls can inflate apparent difficulty.

Dropping inspection obligations or ignoring rack limits produces failures when relaxed plans execute under the actual rules. These are model-mismatch ablations, not strong planners. Raw actions, states, rejections, expansions, timings, and every aggregate are released. Seeds repeat geometry and tie order; cached local timings are not calibrated cross-platform comparisons.

7.2. Historical model integration case

The workbench preserves a version-2.1 SmolVLM-256M-Instruct run on kart-deck-08 (historical public E069, seed 0). All four calls return scene descriptions instead of JSON actions. Each becomes INVALID_ACTION, consumes an operation, and leaves the state unchanged. The controller stops at four calls. This is a failed integration example, not a planning or population result.

The original task, bank hash, image bytes, prompts, raw outputs, parsed actions, feedback, latency, measured MLX token counts, and weight digest are pre-

served. Native stimuli are 1920 pixels wide; this model's processor resizes the longest edge to 512 with splitting disabled. HD display does not imply native-HD inference. Historical E069 is not relabeled as a version-2.2 experiment. An OpenAI-compatible practice adapter and a separate usage-accounted study runner support further collection.

8. Release and limitations

The React workbench exposes seven families, complete previews, ID/source/ split filters, answer submission, model playback, and review-JSON export. Its isolated WebGL2 renderer preserves original geometry, supports free practice orbit and explosion, and shows part details on hover. Native zoom preserves raster evidence. Dataset cards, public JSONL, sample HD inputs, evaluator dependencies, and deployment files are packaged together. No external hosting URL is claimed.

Software validation replays all 377 optimal episodes and 1,952 answers. Audits check sequence multisets, source separation, repair BFS labels, input-pixel differences, budget accounting, missing assignments, overlapping errors, and transcript tampering. Protocol and browser checks cover persistence, observation isolation, and playback. This establishes software behavior, not human usability or real-world mechanical validity.

The principal limits remain structural and empirical. The deterministic bank is tractable with perfect relations. Registration and inventories simplify grounding. Visual-context evidence spans three sources and has a part-count confound; pixel distinction needs empirical answerability testing. Geometry excludes forces, arbitrary poses, rotations, simultaneous subassembly motion, and real rack transfers. Sixteen sources remain below the desired 30–50-source scale. No model shortsightedness, human–model gap, or immunity to all shortcuts has been measured. The next study must report matched static, active, and structured inputs, source-level uncertainty, and count-matched context replication.

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